

Confidence region of the Optimum

An example with one factor

Effect on the optimization

- Let us suppose that the true function is

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon \quad \text{with} \quad \varepsilon \sim N(0, \sigma^2)$$

- You make your experiment and you fit the function

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 \quad \text{and} \quad \Sigma_{\hat{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \hat{\sigma}^2 \quad \text{and} \quad \hat{\sigma}^2 \equiv MS_E$$

- Let us suppose that you want to find the unconstrained minimum of the expected value of the response variable

$$E(y) = E(\hat{y}) \quad \longrightarrow \quad \text{We work on the } \hat{y}(\mathbf{x})$$

- Usually you apply the Necessary&Sufficient conditions to find the minimum

$$\frac{d\hat{y}}{dx} = 0 \text{ and } \frac{d^2\hat{y}}{dx^2} > 0$$

- After some algebraical manipulation

$$\hat{x}_{opt} = -\frac{\hat{\beta}_1}{2\hat{\beta}_2} \text{ and } \hat{\beta}_2 > 0$$

$$\Sigma_{\hat{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \hat{\sigma}^2$$

$$\hat{y}_{opt} = \hat{\beta}_0 - \frac{\hat{\beta}_1^2}{4\hat{\beta}_2}$$

- The problem is to find an interval R_0 such that

$$\Pr[x_{opt} \in R_0] \geq 1 - \alpha$$

Methods

- Analytical
- Monte Carlo

Analytical Method

Consider the estimated model parameters **excluding** the model constant

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\mu}_{\hat{\boldsymbol{\beta}}} = \begin{Bmatrix} \mu_1 \\ \mu_2 \end{Bmatrix} \quad \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$$

Developing in Taylor series $\hat{x}_{opt}$ to the **fourth** term (the expansion to the eighth term changes the figures only to the fourth significant figure) we have

Expected value and Variance (after the Taylor's expansion)

$$E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) = \frac{-\mu_1\mu_2^4 + \rho\mu_2^3\sigma_1\sigma_2 - \mu_1\mu_2^2\sigma_2^2 + 3\rho\mu_2\sigma_1\sigma_2^3 - 3\mu_1\sigma_2^4}{2\mu_2^5}$$

$$V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) = \frac{1}{4\mu_2^{10}}(\mu_2^8\sigma_1^2 - 2\rho\mu_1\mu_2^7\sigma_1\sigma_2 + \mu_1^2\mu_2^6\sigma_2^2 + 3\mu_2^6\sigma_1^2\sigma_2^2 + 5\rho^2\mu_2^6\sigma_1^2\sigma_2^2 - 16\rho\mu_1\mu_2^5\sigma_1\sigma_2^3 + \\ + 8\mu_1^2\mu_2^4\sigma_2^4 + 15\mu_2^4\sigma_1^2\sigma_2^4 + 54\rho^2\mu_2^4\sigma_1^2\sigma_2^4 - 108\rho\mu_1\mu_2^3\sigma_1\sigma_2^5 + 39\mu_1^2\mu_2^2\sigma_2^6 + \\ + 45\mu_2^2\sigma_1^2\sigma_2^6 + 261\rho^2\mu_2^2\sigma_1^2\sigma_2^6 - 402\rho\mu_1\mu_2\sigma_1\sigma_2^7 + 96\mu_1^2\sigma_2^8 + 105\sigma_1^2\sigma_2^8 + 840\rho^2\sigma_1^2\sigma_2^8)$$

Assuming Gaussian distribution we obtain:

$$R_0 = \left[E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) - z_{\alpha/2} \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)}; \quad E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) + z_{\alpha/2} \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)} \right] \quad \text{Or}$$

$$R_0 = \left[E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) - t_{\alpha/2}(N-3) \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)}; \quad E\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right) + t_{\alpha/2}(N-3) \sqrt{V\left(-\frac{\hat{\beta}_1}{2\hat{\beta}_2}\right)} \right]$$

Monte Carlo method

Make the experiment $\longrightarrow \hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2 = MS_E$

$i : 1, N_{Sim}$

$$\hat{\boldsymbol{\beta}}(i) \sim MN\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}} = (\mathbf{X}^T \mathbf{X})^{-1} MS_E\right)$$

$$\hat{x}_{opt}(i) = -\frac{\hat{\beta}_1(i)}{2\hat{\beta}_2(i)}$$

Build the Histogram

Pick up the quantiles 2.5% and 97.5% to have an estimate of the interval of the Optimum point

Example

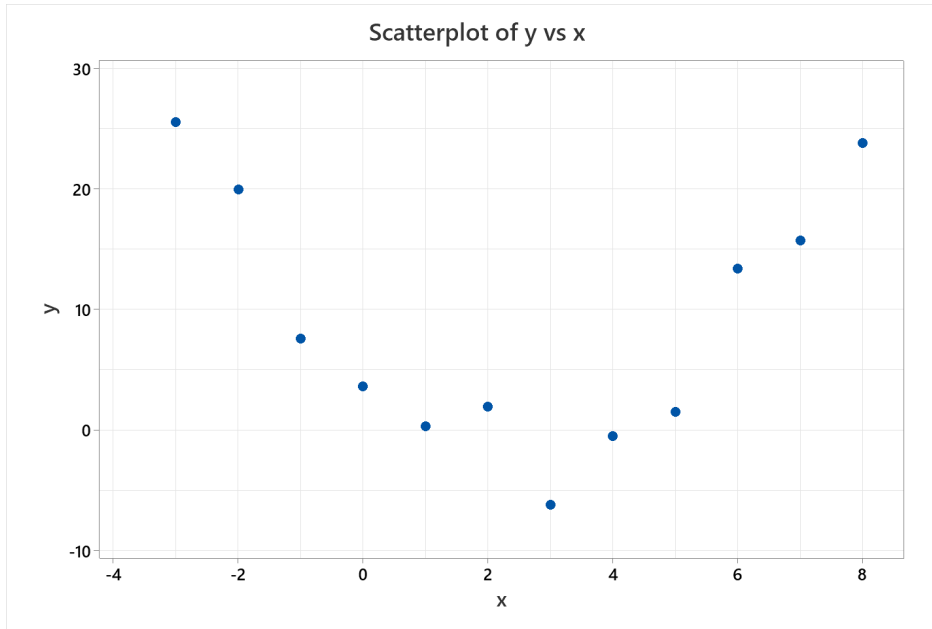
Let us suppose that the unknown true model is

$$y = 4 - 5x + x^2 + \varepsilon \quad \text{with} \quad \varepsilon \sim N(0, 3^2)$$

Assume:

- 12 treatments
- the Experimentation domain is $[-3, 8]$
- the treatments are equidistant: $[-3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8]$
- No replications

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 12 & 30 & 218 \\ 30 & 218 & 1260 \\ 218 & 1260 & 8870 \end{bmatrix} \quad (\mathbf{X}^T \mathbf{X})^{-1} = \begin{pmatrix} 0.1511 & 0.0038 & -0.0043 \\ 0.0038 & 0.0257 & -0.0038 \\ -0.0043 & -0.0038 & 0.0008 \end{pmatrix}$$



The $\mathbf{y}$ vector of the experiment is:

25.56
19.96
7.60
3.63
0.30
1.96
-6.22
-0.53
1.51
13.40
15.75
23.81

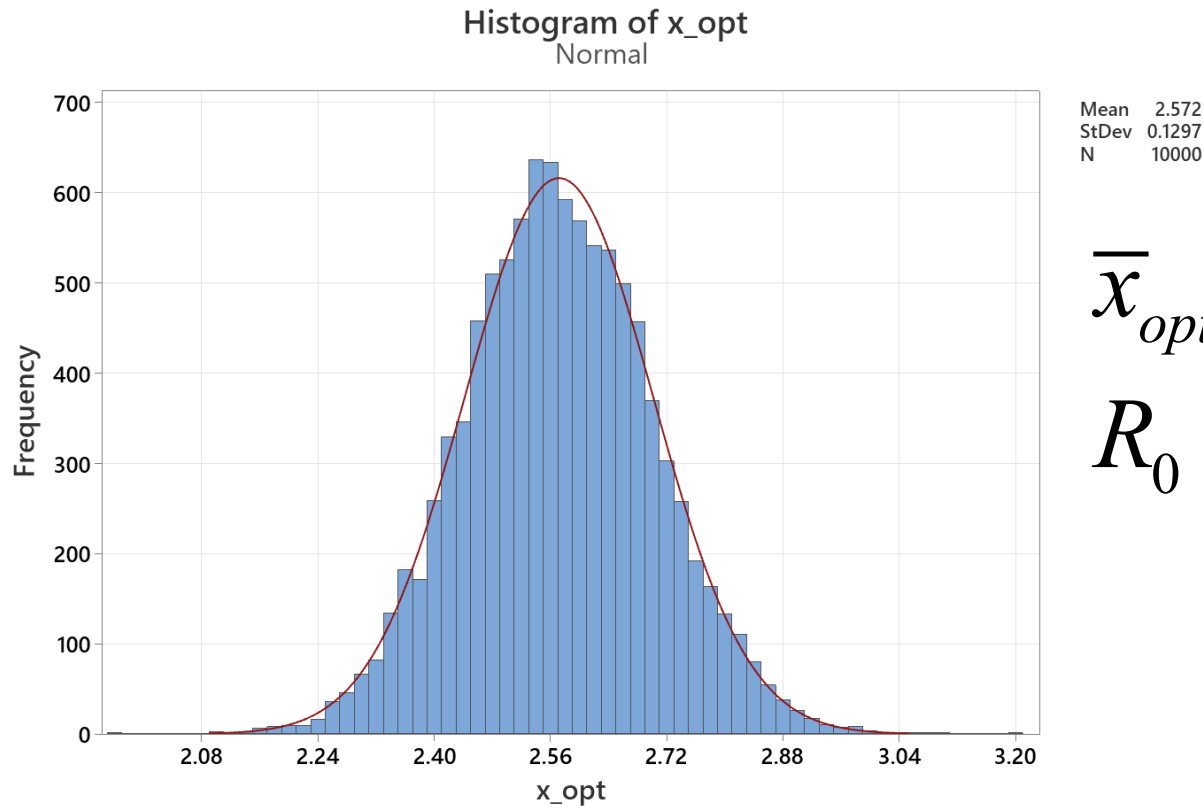
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	2	1140.65	570.32	70.85	0.000
x	1	876.89	876.89	108.93	0.000
x*x	1	1138.15	1138.15	141.38	0.000
Error	9	72.45	8.05		
Total	11	1213.10			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2.83725	94.03%	92.70%	90.15%

We generate the Histogram of the optima and we pick up the quantiles 2.5% and 97.5%



$$\bar{x}_{opt} = 2.572$$

$$R_0 = [2.322; 2.830]$$

If we use the Analytical solution we have $R_0=[2.317; 2.827]$

The true Expected optimum is 2.5

You can play: changing the variance, changing the Design..

Different stochastic problems

$$\text{Min}_x \hat{y}(x)$$

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s.t. $x \in [a, b]$ a and b deterministic constant

$$\text{Min}_x E\left(y\left(x, \hat{T}(x)\right)\right)$$

$$y = \frac{\alpha}{v} \left(1 + \frac{\varphi}{\hat{T}(v)} \right)$$

$$\text{s.t. } \Pr\left\{\hat{P}(x) \leq P_0\right\} = 1 - \alpha$$

$$\hat{P}(v) = \eta \hat{F}v$$

And many other problems